

Aniol Escofet

Software & Electronic Engineer



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📠 Aniol Escofet 📞 +34 688811982

📅 13/11/2001 🚗 A1, B

📄 PROFILE

Passionate and innovative automotive engineer specializing in vehicle dynamics, control systems, and performance optimization. Experienced in simulation, data analysis, and race engineering support, with hands-on involvement in both professional motorsport and research environments. Currently working at Stark Future S.L. and pursuing a Master's in Industrial Engineering. Eager to apply skills to maximize car and driver performance within a competitive Formula 1 environment.

🌐 LANGUAGES

Català - Native
Espanyol - Native
English - B2 Grade B

🛠 TECHNICAL SKILLS

MATLAB	
Simulink	
Python	
C++	
WinTax	
GEMS	
Ecotrons	
IPG	
VI-Grade	
Git	
Solid Works	
Excel	

📁 PROFESSIONAL EXPERIENCE

STARK Future S.L. 🔗

Junior Power Electronics R&D
01/2025 – present | Sant Boi de Llobregat

- Built and validated a full-vehicle 1D simulation model including battery, inverter, motor, gearbox, chassis and environment.
- Conducted system-level validation through simulation and physical testing.
- Performed system-level trade-off analysis and testing on battery capacity, torque, gearing, and power output combinations to maximize performance across MX, SM and street motorcycles.
- Developed and validated traction control algorithms and throttle profiles for racing applications, focusing on drivability, wheel slip management, and performance optimization under varying grip conditions.

Cupra Racing 🔗

Software & Electronic Engineer Intern
09/2023 – 12/2024 | Martorell

- Developed control systems and advanced algorithms for vehicle dynamics using MATLAB and Simulink.
- Conducted simulation and validation campaigns (MiL, SiL, HiL, DiL) to optimize vehicle performance.
- Led vehicle-level testing for performance and durability, analyzing track data to improve car and driver behavior.
- Supported race operations by integrating and validating prototype systems, ensuring reliability under competition conditions.
- Optimized CAN, GPS, and IMU communication systems for accurate performance monitoring.
- Coordinated with multidisciplinary teams to analyze test outcomes and implement innovative solutions for system improvement.

BCNeMotorsport 🔗

Vehicle Controls
09/2021 – 09/2024 | Barcelona

- Designed and implemented control systems to solve complex problems and enhance system performance of a Formula Student single-seater.
- Developed and applied both dynamic and kinematic vehicle models for simulation and validation.
- Directed test planning, execution, and data analysis, iterating designs based on performance outcomes.
- Analyzed driver and vehicle data to support performance improvements and race strategy decisions.
- Collaborated with multidisciplinary teams to integrate mechanical and electronic subsystems.
- Experience in understanding and implementing vehicle dynamics.

🎓 EDUCATION

Master's Degree in Mechanical Engineering - Automotive & Motorsport

Politecnico di Milano - Campus Bovisa 🔗
2025 – present | Milan

- Focused on motorsport engineering with courses in Vehicle Dynamics & Control, Advanced Tyre Modelling, Human-in-the-Loop Simulation, Motorsport Engineering, Hybrid & Electric Vehicles, and Optimal Vehicle Design.

Master's Degree in Industrial Engineering - Mechanical

Universitat Politècnica de Catalunya - UPC
2024 – present | Barcelona

- Focus on control systems, robotics, PLC programming, and automation software/hardware integration.
- Coursework on internal combustion engines, mechanical components and electronic integration.

Bachelor's Degree in Industrial Engineering

Universitat Politècnica de Catalunya - UPC
2019 – 2024 | Barcelona

- Strong foundation in mechanical design, data analysis, control systems, and engineering programming (Python, MATLAB).
- Thesis: Model-Based Torque Vectoring for a 4WD Formula Student Vehicle 🔗